

LESSON 1.6

FINDING THE PERIMETER AND AREA OF RECTANGLES



LESSON INTRODUCTION

MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.

LEARNING TARGETS

- I can find the perimeter and area of a rectangle with fractional side lengths.
- I can find the perimeter and area of a rectangle with decimal side lengths.

BENCHMARK COHERENCE

BUILDING ON

MA.5.AR.1.2 Solve real-world problems involving the addition, subtraction, or multiplication of fractions, including mixed numbers and fractions greater than one.

WORKING TOWARD

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.

ESSENTIAL QUESTIONS

- What is the difference between perimeter and area?
- How can you find the perimeter and area of rectangles?
- How does the area model relate to area of a rectangle?

LESSON NARRATIVE

Students use understanding of operations with fractions to find the perimeter and area of given shapes represented on and off a grid, using models, and through formulas. Students compare and contrast approaches to perimeter and area using gridlines, including intentional problem-solving contexts that include fractional side lengths.

COMPONENT SUMMARY

1.6.1 WARM-UP (~5 MIN.)

Using gridlines to form rectangles

1.6.2 GUIDED INSTRUCTION (25 MIN.)

Using gridlines to explore area and perimeter

1.6.3 COLLABORATION (10 MIN.)

Comparing perimeter and area in problem-solving contexts

1.6.4 YOUR TURN! (5 MIN.)

Students find the perimeter and area of a composite shape in real-world context

1.6.5 WRAP-UP (~5 MIN.)

Error analysis on the use of formulas for perimeter and area

RESOURCES

GLOSSARY

Key Terms:

- Perimeter (of a polygon)
- Area of a rectangle

Additional Terms:

- N/A

MATERIALS

- (Optional) 1.6.1 Warm-Up video
- (Optional) Highlighters or markers

ADDITIONAL PREPARATION

- For 1.6.2 Guided Instruction, create an optional anchor chart on the difference between area and perimeter and/or the definition for multiplication properties.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse perimeter and area.
- Students may need help connecting area and perimeter to operations with fractions from previous lessons in this unit.

THINGS TO CONSIDER

- Strengthen awareness of the difference between area and perimeter by giving students a variety of contexts that would require finding both area and perimeter, then ask students how we would find the answer for each.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

This lesson provides opportunities to engage in metacognitive practices when connecting previously learned skills in fraction operations and clarifying real-world problems and operations/steps needed to solve. Throughout 1.6.2 Guided Instruction, use Think Aloud routines to help students compare and connect how finding the area of a rectangle with fractional side lengths is like the area model with fraction multiplication.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

1.6.2 Guided Instruction and 1.6.3 Collaboration position students to use fractions with operations and knowledge of area and perimeter to solve real-world problems. Students use models and methods to understand, represent, and solve for both area and perimeter diagrams within the real-world contexts.

DIFFERENTIATE INSTRUCTION

Notice the use of a variety of strategies for performing operations with fractional side lengths. Provide space and opportunity for students to independently and collectively analyze strategies and approaches with both gridlines and area models. Students can leverage On-Ramp to 6th Grade videos on operations with fractions including multiplication and division. However, the lesson is intentionally constructed to position students to apply strategies like area model and gridlines for fractional side lengths.

1.6.1 WARM-UP: 24 LINES

Component Overview: Read the prompt as a class, then have students independently complete question 1. If possible, play the video and ask students what they notice.

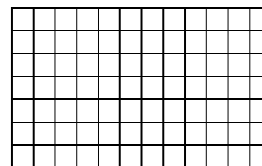
TEACHER MOVES

Before students build the rectangle for question 1, consider discussing how counting gridlines is different from counting boxes. This understanding will be very important as students begin using the formula to find area and perimeter and will need to count distance between intersections on the grid, not boxes, to determine side lengths.



1.6.1 Warm-Up: 24 Lines

Use only 24 of the gridlines to draw a rectangle.



- How many different rectangles do you think can be drawn using only 24 of the gridlines?

Answers will vary.

- b. What is the perimeter of the rectangle?

14 units

- c. If each gridline has a length of 1 centimeter (cm), write the perimeter of the figure using units.

14 cm

Area is the number of unit squares that are needed to cover a space. One way to find the area of the above figure would be to count the unit squares in the figure.

5. Discuss with your partner why counting unit squares to find the area of the trapezoid or triangle would be difficult. **Exact length of diagonal sides can't be counted.**

Some polygons have a formula that can be used to find the area. Other kinds of figures have different formulas for area.



Area of a rectangle

$A = l \cdot w$ where l is the length and w is the width

6. Consider the rectangle on the grid.

- a. If each gridline has a length of 1 centimeter (cm), what is the unit for the area of one square on the grid?

1 cm² or 1 square centimeter

- b. What is the area of the rectangle? Be sure to include units.

32 cm²



1.6.2 GUIDED INSTRUCTION: FINDING PERIMETER AND AREA

Component Overview: During this Guided Instruction, students first encounter the idea of perimeter with a rectangle, a trapezoid, and a triangle on a grid to identify the type of operation to be used. Then instruction relates the concept of perimeter to area, using the same figures on a grid, ending with samples of problem-solving strategies and exploration of what units are appropriate for each.

DIFFERENTIATED INSTRUCTION

For question 2, if students are having trouble seeing that a diagonal is longer than a horizontal and therefore is a different measurement, consider demonstrating how the measurement of the height or length of a straight vertical or horizontal line is shorter than a slightly diagonal version.

Students may benefit from measuring from one point on a tile floor to another point on a diagonal by counting the number of steps or using a tape measure and comparing that to the distance using the grid of the tiles.

TEACHER MOVES

Compare and Connect

For questions 2 and 3, consider having students share why finding the perimeter of a triangle is different from finding the perimeter of a rectangular figure. This may help them distinguish figures that can use the formula for perimeter while also solidifying the concept of perimeter of a polygon. Questions might include:

- Is it possible to use multiplication to find perimeter? Why or why not?
- How is finding the perimeter of the green triangle different from the orange rectangle?
- What makes a polygon different from a rectangle?
- Do you think the formula for perimeter of a rectangle will work for a triangle? Why or why not?

DIFFERENTIATED INSTRUCTION

To support visual and kinesthetic learning approaches prompt students to use different colored highlighters or markers to underline the definition of perimeter and area. Then, using the same color for perimeter, have students trace the perimeter of each shape in question 5, and use the same color for area to shade the area.

1.6.2 GUIDED INSTRUCTION: FINDING PERIMETER AND AREA (CONTINUED)

TEACHER MOVES

Compare and Connect

Notice the concept of area is considered after the concept of perimeter, which is purposeful, as students will have already considered the side lengths of these same figures. Consider having students compare and connect the two measurements by explaining what the perimeter and area are for one of the shapes in question 5.

ENRICHMENT

To help students understand what is a square unit and why units for area are written as units squared, consider asking students to describe how counting the squares in the rectangle on the grid also results in the same value as using the formula $A = l \cdot w$ to find the area. Students may not understand why units can be written as cm^2 so use of this abbreviation may need to wait until Unit 8. In grade 5, student responses for units when finding area or perimeter were in word form.

TEACHER MOVES

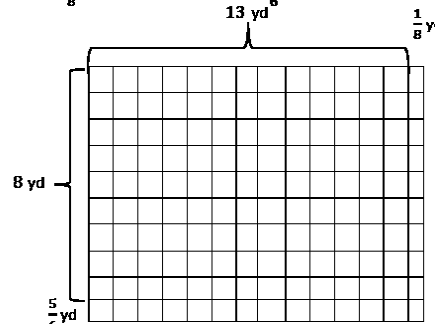
Compare and Connect

Consider prompting students to try individually identifying the similarities and differences between the two strategies and/or the representation for Rashona and Naira, then regrouping to share ideas in a class discussion. Encourage students to annotate the models and/or write observations or questions to share with the class.

DIFFERENTIATED INSTRUCTION

On question 7a, prompt students to remember multiple approaches to solving fraction problems, including using improper and equivalent fractions to solve, then put their final answer in the proper mixed-number format and use units to describe the area. Promoting alternate approaches as an entry point can foster flexibility as students approach problem solving with fractions.

7. Zoo Miami is the largest zoo in Florida and home to over 3,000 animals. In the Florida Mission Everglades section, the zoo houses seven different species of animals that are native to the state of Florida. One of the animals is the American black bear. The zookeeper wants to build the black bears a rectangular pool in their exhibit that is $13\frac{1}{8}$ yards (yd) wide by $8\frac{5}{6}$ yd long, as shown.



The table shows the strategies of two different students for finding the area of the pool for the American black bears.

Rashona's Strategy	Naira's Strategy
I sketched the area and divided it into four sections. I found the area of each section. Then, I added all of the areas.	There are $8\frac{5}{6}$ rows of $13\frac{1}{8}$ squares so I found $8\frac{5}{6} \times 13\frac{1}{8}$.

- a. Use Rashona's strategy to find the area of the pool for the American black bears.

$8 \cdot 13 = 104$	$8 \cdot \frac{1}{8} = 1$
$\frac{5}{6} \cdot 13 = 10\frac{5}{6}$	$\frac{5}{6} \cdot \frac{1}{8} = \frac{5}{48}$

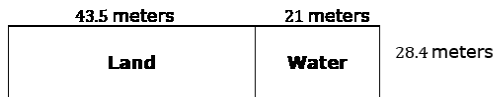
$$\text{Area of the pool} = 115\frac{15}{16} \text{ yd}^2$$

- b. Use Naira's strategy to find the area of the pool for the American Black Bears.

$$8\frac{5}{6} \cdot 13\frac{1}{8} = \frac{53}{6} \cdot \frac{105}{8} = 115\frac{15}{16} \text{ yd}^2$$

- c. What are the units for the area of the pool?
Square yards or yd²
- d. How much fencing is needed to enclose the pool area?
 $43\frac{11}{12}$ yards
- e. What are the units for the amount of fencing for the pool?
yards
8. Discuss with your partner the difference between finding the perimeter of a rectangle and finding the area of a rectangle. Write down a strategy that you can use to remember the difference.
Answers will vary. Perimeter uses addition and area uses multiplication. Perimeter is the fence while area is the yard.

9. The American alligator is also featured in the Florida Mission Everglades area. A diagram of the alligator enclosure is shown.



- a. What is the area of the water enclosure? Be sure to include the units.
596.4 meters²
- b. Nolan's work for finding the area of water enclosure is shown below.

21 · 28
21 · 0.4

21(28 + 0.4)

With your partner, discuss which property Nolan can use to find $21(28 + 0.4)$. Name or describe the property.

Distributive property or multiplication property

- c. What is the area of the alligator enclosure? Be sure to include the units.
1,831.8 meters²
- d. Work with your partner to determine the total amount of fencing needed for the alligator enclosure. Show your work or strategies you used.
185.8 meters

1.6.2 GUIDED INSTRUCTION: FINDING PERIMETER AND AREA (CONTINUED)

ENRICHMENT

For question 9, position students to understand the representation by asking:

- How many parts are in the Mission Everglades alligator area?
- How can you use the measurements of sides you know to find the measurement of sides that you do not know?

1.6.3 COLLABORATION: PERIMETER AND AREA WITH FRACTIONAL AND DECIMAL SIDE LENGTHS

Component Overview: Working in partners, students synthesize their understandings of perimeter and area by determining which concept is required and the appropriate unit in real-world situations.

DIFFERENTIATED INSTRUCTION

Encourage collaboration by having one partner draw a diagram of the situation and the other shade or trace the part of the diagram that they need to find (representing either area or perimeter) for the real-world situation. Have partners switch roles for each situation so both partners get practice creating accurate models and identifying whether area or perimeter is being found.

TEACHER MOVES

Note that question 2 revisits guidance options in 1.6.2 Guided Instruction. Use question 2 as an opportunity for students to rephrase the reasoning in their own words. If students have difficulty explaining the connection, prompt them to consider the part of the grid they are focusing on—the lines or the area inside the lines.

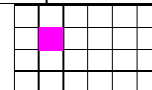


1.6.3 Collaboration: Perimeter and Area with Fractional and Decimal Side Lengths

1. Work with your partner to complete the table. Choose whether the real-world situation represents area or perimeter, then write the appropriate units for the situation.

Real-World Situation	Area or Perimeter	Units
The amount of paint needed to paint a wall that is 6 meters (m) long and $2\frac{5}{8}$ m high.	Area	Square meter or m^2
The amount of fencing needed to enclose a backyard that is $\frac{1}{5}$ mile (mi) long and $\frac{1}{20}$ mi wide.	Perimeter	Miles (mi)
The number of tiles needed to put a border on a kitchen floor that is $4\frac{1}{6}$ yards (yd) long and $3\frac{3}{4}$ yd wide.	Perimeter	Yards (yd)
The number of tiles needed to cover a patio that is 775.8 centimeters (cm) long and 582.1 cm wide.	Area	Square centimeters or cm^2

2. Work with your partner to explain how the square in the grid relates to the units of area being "square units."



Answers will vary. If 1 gridline represents 1 unit, then the shaded square has side lengths of 1 unit. Therefore, the area of the shaded square is (1 unit $\times$ 1 unit = 1 square unit).

1.6.4 YOUR TURN!

Component Overview: In this independent activity, students use a provided diagram to calculate both the area and perimeter of a polygon using yards and fractions. Students should be expected to show their work and use appropriate units based on the context of the problem.

DIFFERENTIATED INSTRUCTION

For added visual support for finding missing side lengths, have students trace sides that they know the measurements of using a colored pencil or highlighter. Students can then see how sides can be used to find the missing values.

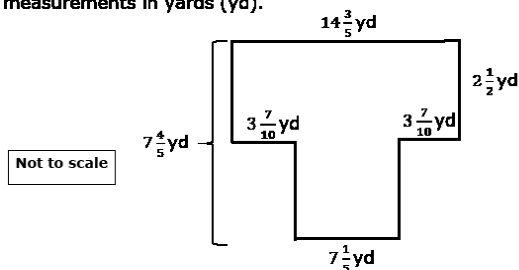
TEACHER MOVES

Prompt students to explain their steps and formulas used to solve for perimeter and area, reflecting on the decisions they made to find the missing side lengths.



1.6.4 Your Turn!

The giant anteaters' habitat is shown in the diagram with measurements in yards (yd).

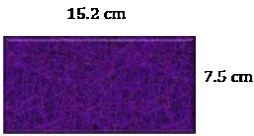


1. The zoo wants to build a walking path around the outside of the enclosure. How long will the walking path be?
 $44\frac{4}{5}$ yards
2. Giant anteaters prefer to live in an area covered with grass and trees. How large is the area that needs to be covered with grass and trees?
 $74\frac{33}{50}$ square yards



1.6.5 Wrap-Up: Error Analysis

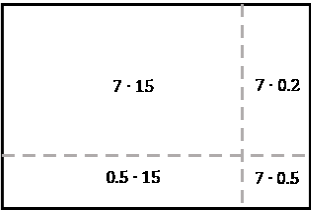
1. Rodrick was designing a kitchen. One of the tiles he wanted to use is shown with measurements in centimeters (cm). Rodrick represents the tile's perimeter in different ways.



- a. Determine which representations are correct.

Perimeter	Correct?
$P = 15.2 + 7.5$	☐ yes ☐ no
$P = 2(15.2 + 7.5)$	☐ yes ☐ no
$P = 15.2 + 7.5 + 15.2 + 7.5$	☐ yes ☐ no

- b. Rodrick represents the area of the tile using the diagram shown.



Rodrick made a mistake when labeling the smaller areas. Correct Rodrick's mistake.
7 · 0.5 should be 0.5 · 0.2

1.6.5 WRAP-UP: ERROR ANALYSIS

Component Overview: This Wrap-Up assesses students' abilities to identify whether a given perimeter expression is appropriate and what mistake was made when calculating area of a rectangle. Have students individually complete this work as a formative assessment of the lesson's learning targets. Students should **not** use a calculator to complete the Wrap-Up.

DIFFERENTIATED INSTRUCTION

For question 1a, consider giving students the opportunity to verbalize their reasoning for why the representations are correct or incorrect. This auditory approach may serve as a more comprehensive entry point for some students who prefer to explain verbally rather than in written form.